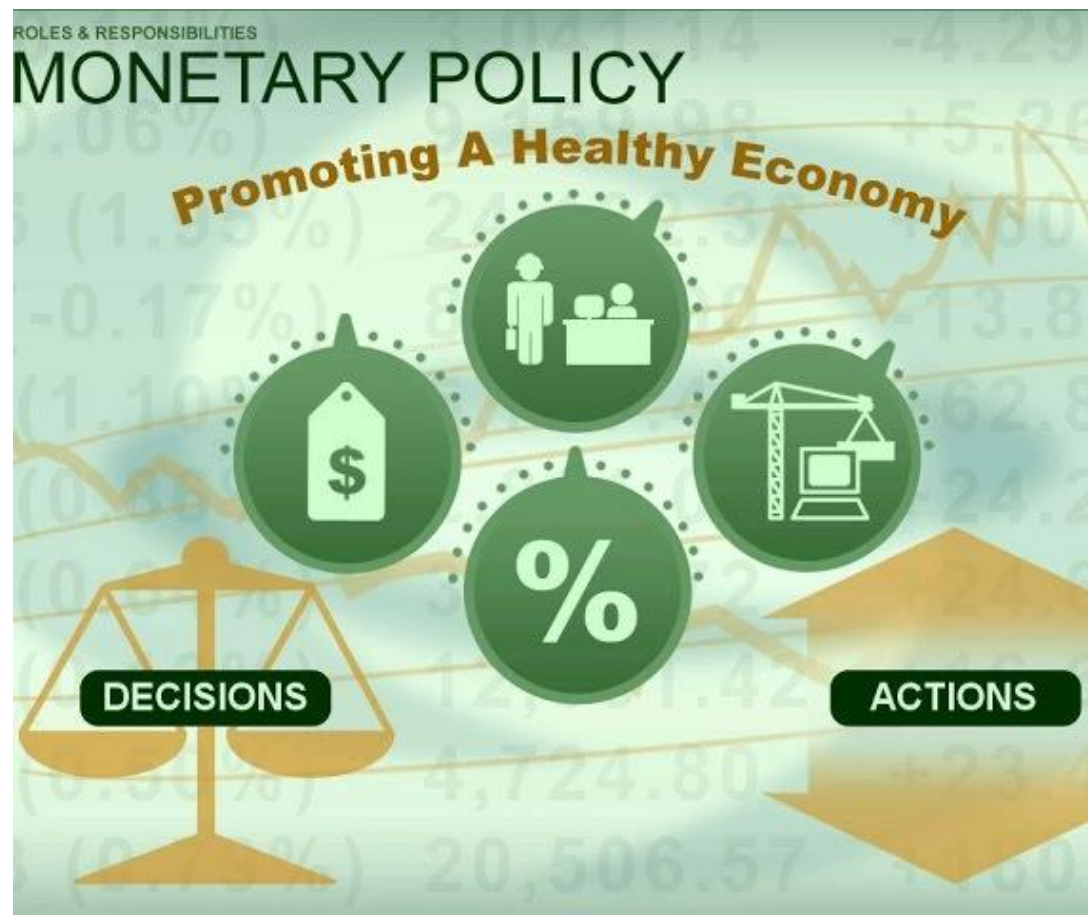


Monetary Policy: Goals, Frameworks, and Tools



- Acknowledgment: This part was built based on Prof. Wook Sohn and Dr. Wytone Jombo lecture materials.

Role of the Central Bank



Why Monetary Policy Matters

- Stabilizes the economy over the business cycle
 - Keeps inflation low and predictable
 - Supports employment and financial stability
 - Provides the 'rules of the game' for firms and households
- “Monetary policy is action that a country's central bank or government can take to influence how much money is in the economy and how much it costs to borrow” (Bank of England).

Friedman's Insight

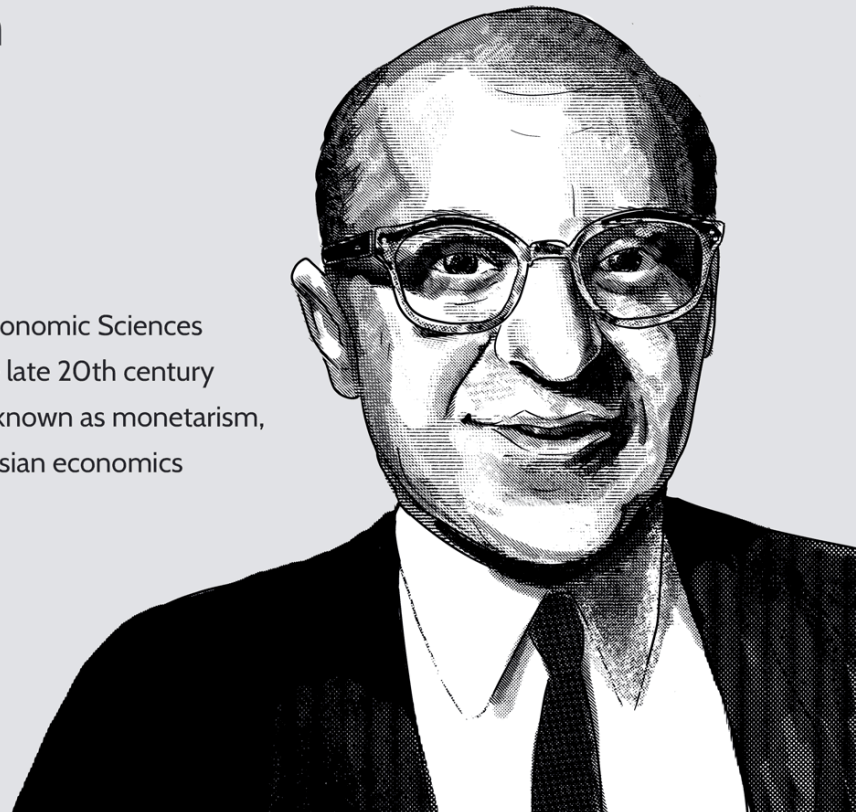
Milton Friedman

Born: July 31, 1912

Economist

- 1976 Nobel Prize Recipient in Economic Sciences
- A leading economic voice of the late 20th century
- His economic theories became known as monetarism, and refute key aspects of Keynesian economics

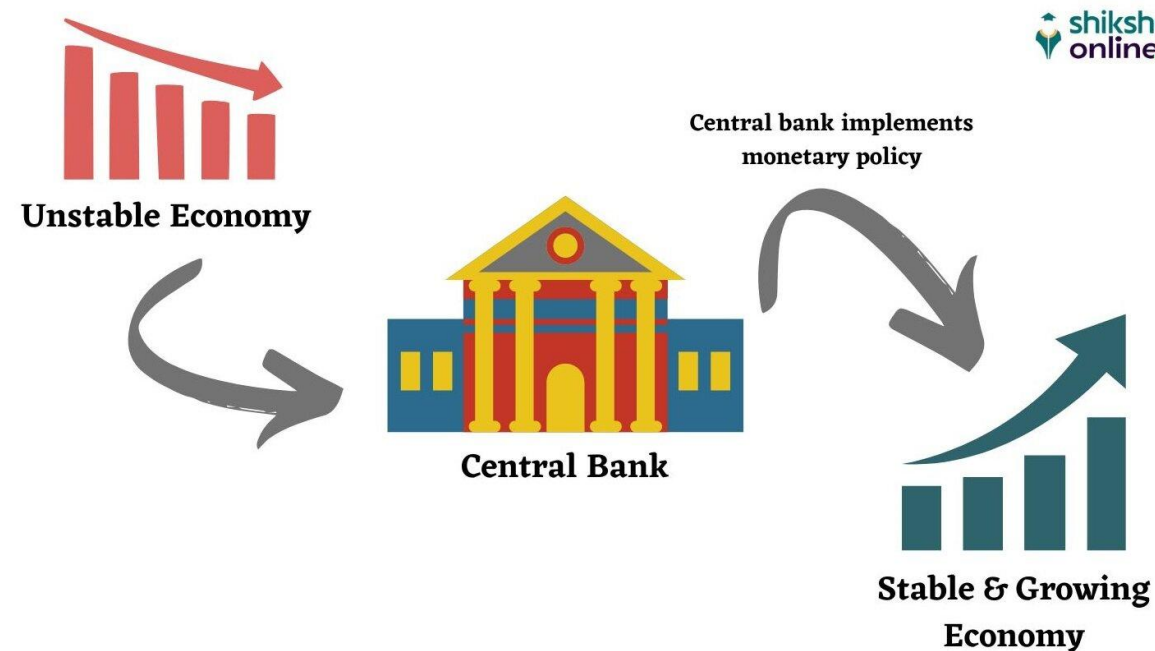
 Investopedia



- "Monetary policy can prevent money itself from being a major source of economic disturbances ... and provide a stable background for the economy." – Milton Friedman (1968)

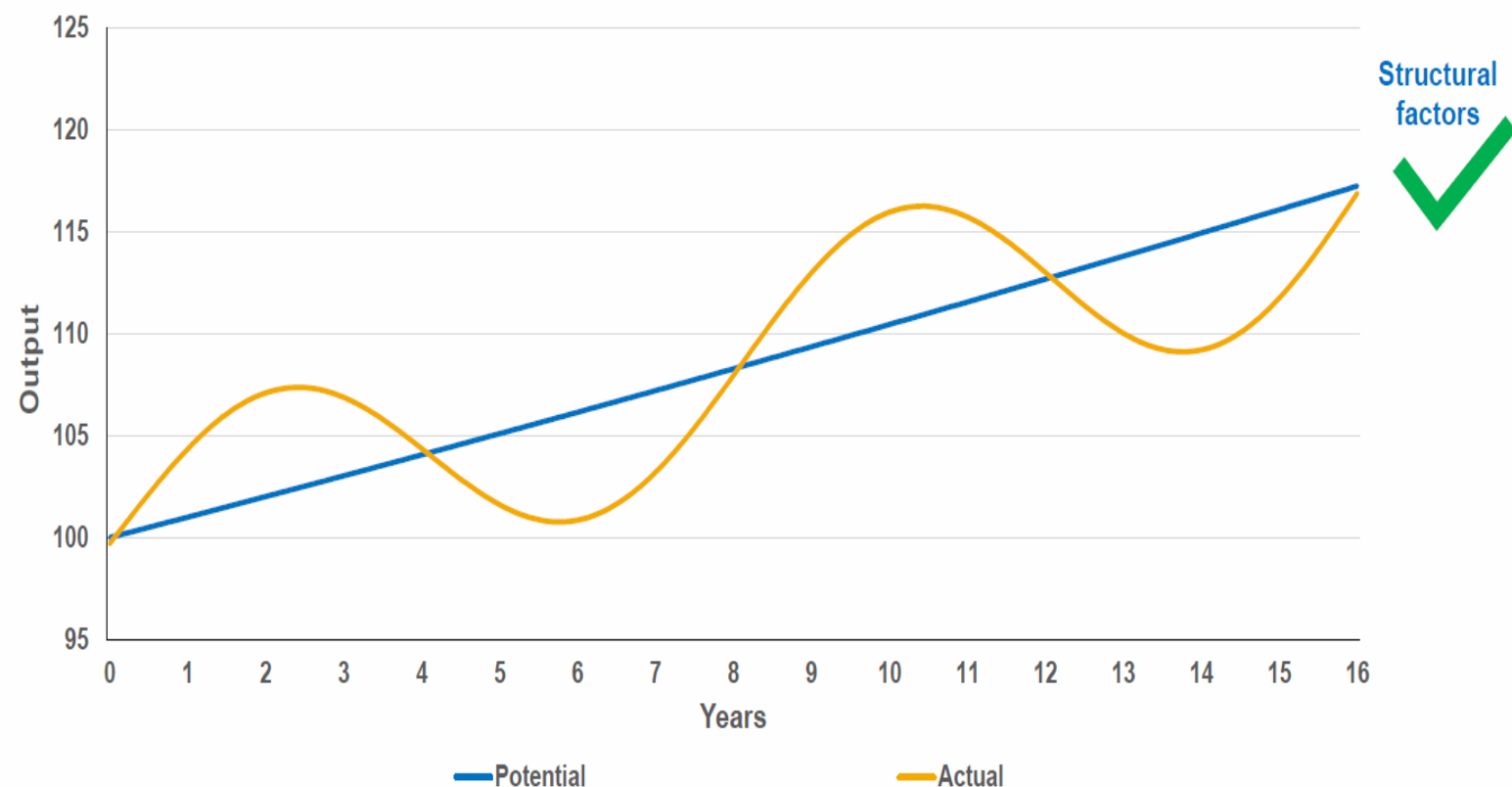
Goals of Monetary Policy

- Price stability (main anchor)
- Economic growth / employment
- Financial stability:
 - Interest rate stability
 - FX stability
 - Market stability



Stylized Economy

- Monetary policy cannot change long-run potential output (set by labor, capital, technology). But it can smooth the cycle around potential



Trend level of output is determined by structural factors e.g. labor, capital, technology, and not MP.

Transmission Channels

How does a policy rate cut affect the real economy?

1. Interest rate channel

2. Expectations

3. Asset prices

4. Credit

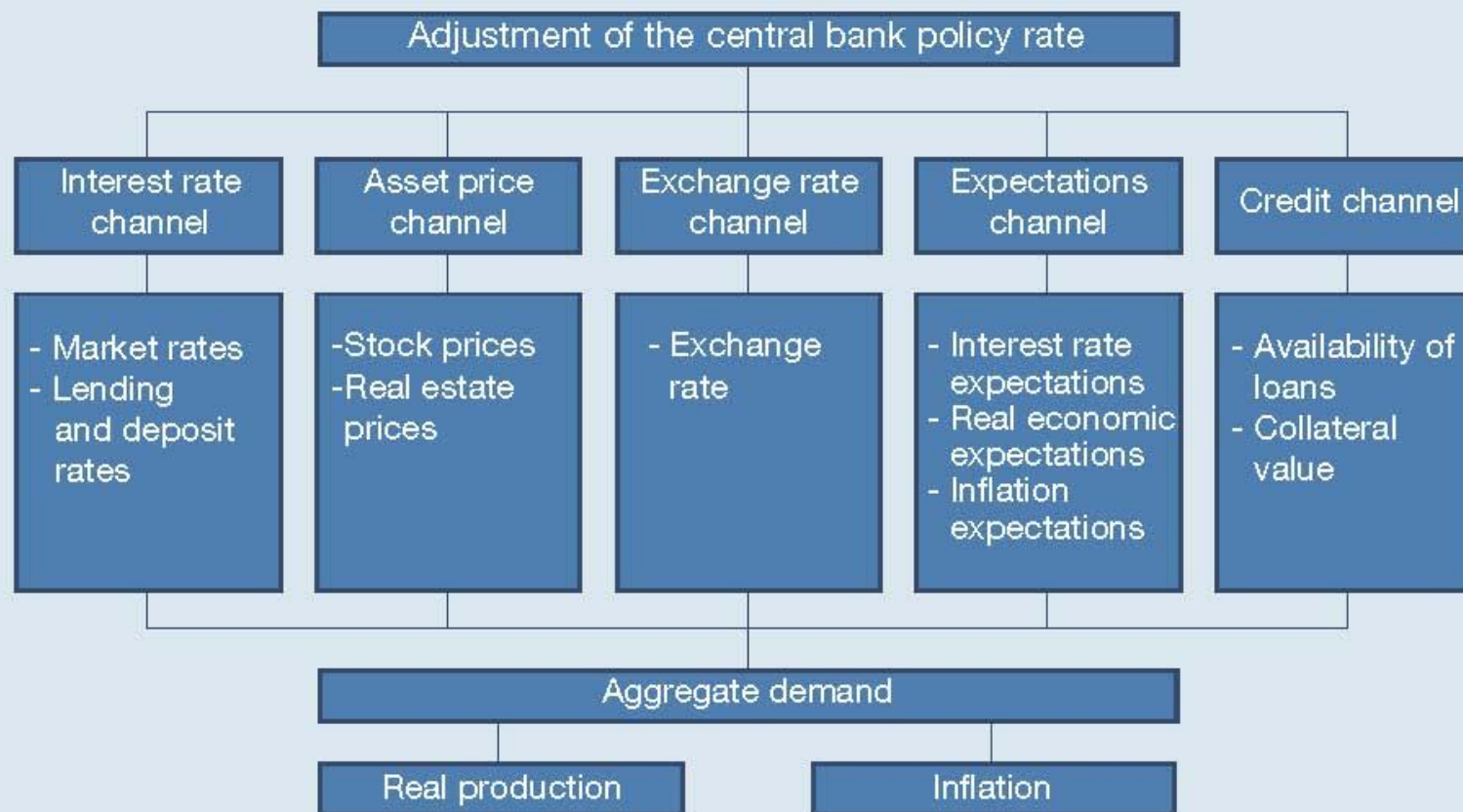
5. Exchange rate

Objectives

Transmission Channels

<Figure III-7>

Transmission channels of monetary policy



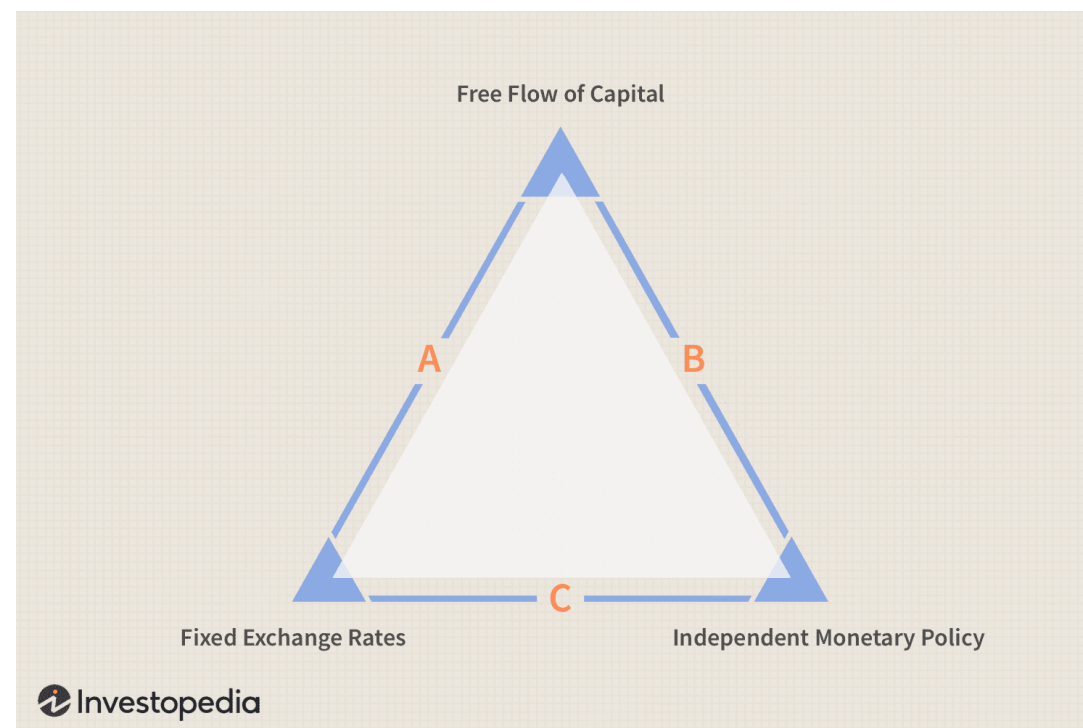
The Monetary Policy Frameworks

SUMMARY	TABLE 1 Advantages and Disadvantages of Different Monetary Policy Strategies		
	Monetary Targeting	Inflation Targeting	Implicit Nominal Anchor
Advantages	Immediate signal on achievement of target	Simplicity and clarity of target Does not rely on stable money–inflation relationship Increased accountability of central bank Reduced effects of inflationary shocks	Does not rely on stable money–inflation relationship Demonstrated success in United States
Disadvantages	Relies on stable money–inflation relationship	Delayed signal about achievement of target Could impose rigid rule (though has not in practice) Larger output fluctuations if sole focus on inflation (though not in practice)	Lack of transparency Success depends on individuals in charge Low accountability

- Exchange rate targeting
- Monetary targeting
- Inflation targeting
- Price-level targeting
- Nominal GDP targeting
- Hybrids ('fear of floating')

The Trilemma

- A country cannot have simultaneously:
 - Free capital mobility
 - Fixed exchange rate
 - Independent monetary policy
- Must choose 2 out of 3



Inflation Targeting

- Publicly announced inflation target (e.g. 2%)
- CB uses policy rate as main tool
- Anchors expectations → stabilizes inflation
- Transparency & accountability



Real-World Example

- New Zealand, UK, Canada
- Early adopters of inflation targeting
- Credibility gains once targets announced

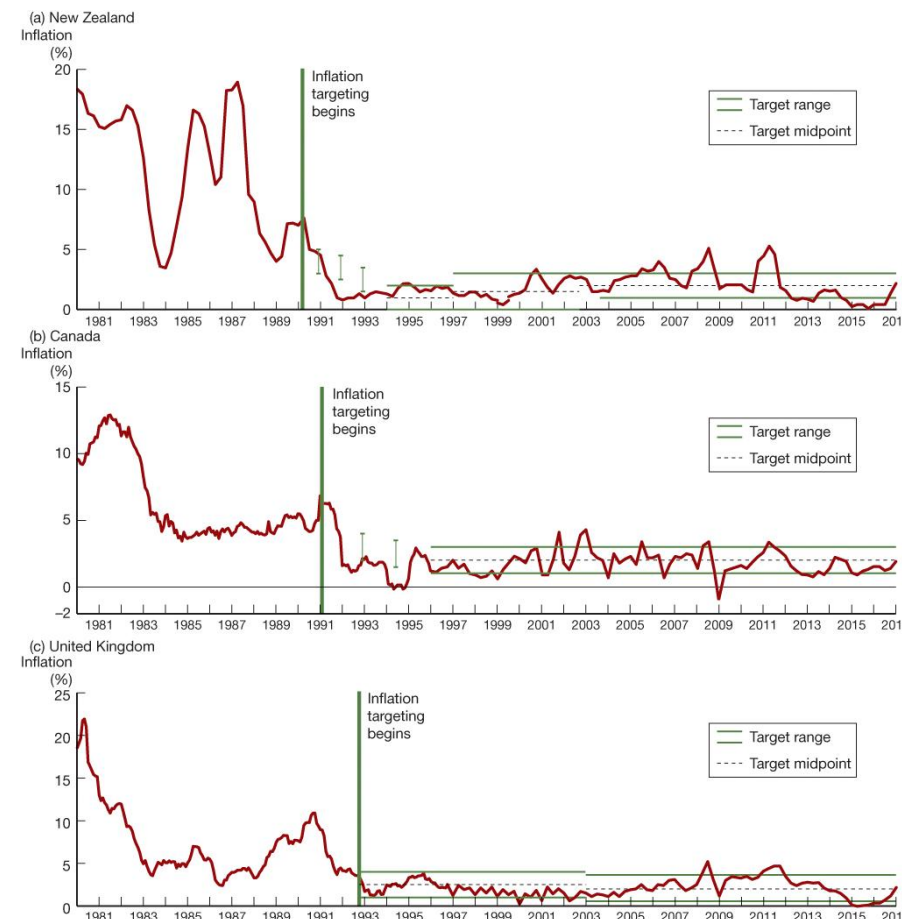
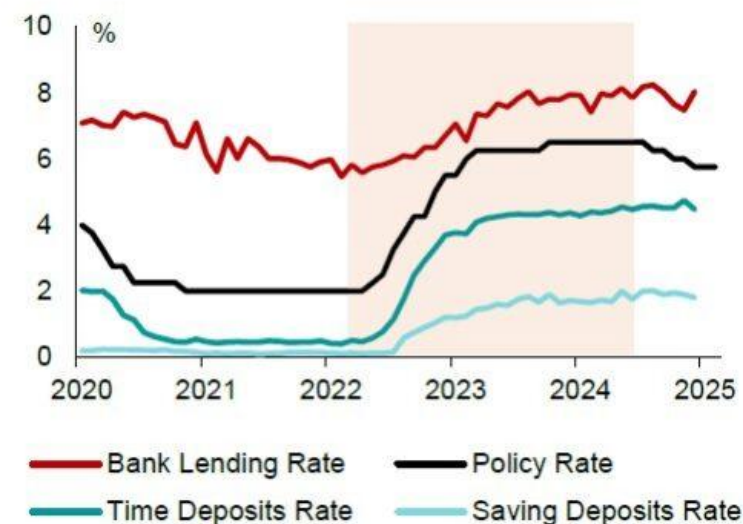


Figure 1 Inflation Rates and Inflation Targets for New Zealand, Canada, and the United Kingdom, 1980–2017

Policy Rate Pass-Through

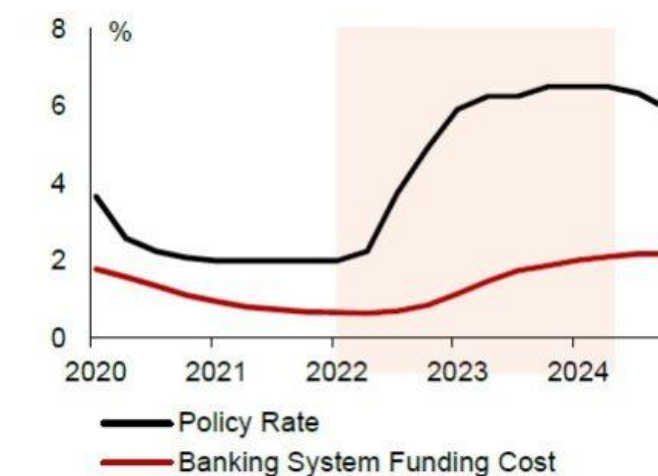
- - CB sets policy rate (wholesale)
- - Banks set lending rate (retail, with markup)
- - During crises, pass-through weakens (credit crunch)

Figure 1. Philippines Interest Rates



Source: BSP; CEIC

Figure 2. Average Bank Funding Cost



Source: BSP; CEIC

Conventional vs unconventional monetary policy

Conventional Policy Tools



Policy Interest Rate

Rate at which banks park their funds with the central bank (and vice versa)



Money Supply

Central banks control monetary base of an economy through buying/selling government securities and printing more money



Unconventional Policy Tools



Quantitative Easing

Purchasing assets (both govt and private) to increase money supply in the economy



Forward Guidance

Central bank commits to a stance of monetary policy until a specific point in time



Other policy tools

Negative interest rates, Helicopter money, adjustments to market operations etc.

Case Study: Global Financial Crisis

- - Fed cut rates from 5.25% → 0%
- - Banks raised markups → lending froze
- - QE introduced to restore credit flow
- - IS–PC–MR: IS stuck left, MR ineffective until QE

Case Study: COVID-19 (2020)

- - IS collapse (demand shock) + PC shift up (supply shock)
- - Policy rate cuts + QE
- - Forward guidance: 'Rates low for long'
- - Stabilized markets, anchored expectations

Central Bank Independence

- - Importance:
 - • Long horizon, free from politics
 - • Independence → lower inflation
- - Risks:
 - • Following markets too closely
 - • Losing long-run anchor

Takeaways

- - MP goals: inflation, growth, financial stability
- - Transmission: via i , expectations, credit, assets, FX
- - Frameworks: monetary targeting → inflation targeting → UMP
- - IS–PC–MR helps analyze crises

Assignment: Exploring Your Central Bank

- Task: Visit the official website of your country's central bank.
- Write a short report (~1 page) covering:
 - Monetary policy objectives
 - Main policy tools
 - Policy framework/regime (e.g., inflation targeting, exchange rate targeting)
 - Any recent policy decisions or statements
- Submission: Upload your report to Cyber Campus.

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macroeconomics

INSTITUTIONS, INSTABILITY, AND INEQUALITY

WENDY CARLIN • DAVID SOSKICE

Module code

Chapter 6: Monetary Policy

Objectives:

By the end of this chapter, students should understand the following:

- The evolution of CB monetary-targeting to inflation-targeting
- The modern *MP* framework and modelling how i is set
- Difficulties of dealing with asset price bubbles
- Use of Quantitative Easing and its possible dangers
- Relationship between expansion of CB balance sheets and inflation

The economy's nominal anchor: money supply

In the early 1970s, the Bretton Woods system of fixed exchange rates came to an end. National authorities chose a new nominal anchor: monetary targeting

- Without fully flexible wages and prices and rational expectations, two conditions must hold for monetary targeting to be successful in controlling inflation:
 - CB must be able to control the chosen monetary aggregate
 - The relationship between inflation and the targeted monetary aggregate must be strong
- Yet those conditions failed to hold in countries like the UK and hence monetary targeting was undermined
 - Goodhart's law prevents CB from controlling monetary aggregate
 - The relationship between narrow money that the CB can control, and inflation is weak

The economy's nominal anchor: the Thatcher experiment

When Margaret Thatcher came to power in 1979, she focused on inflation reduction

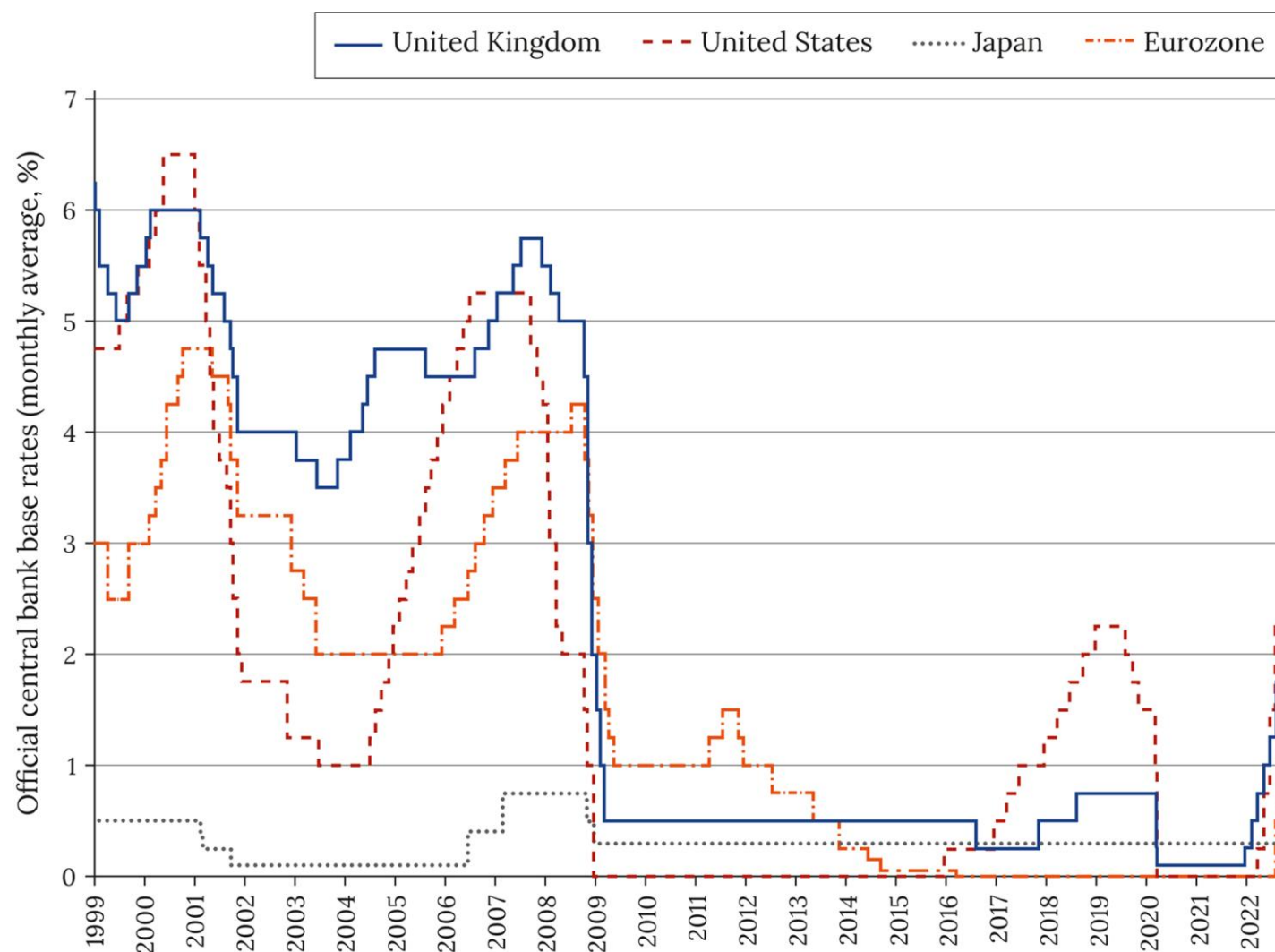
- Failure in Monetary targeting: CB uses M0 to target M3, but relationship broke down → M0 growth > target, unemployment cost of disinflation is very high
- Monetary targeting to anchor π was flawed → Monetarism was questioned → inflation-targeting monetary policy framework emerged

	1980	1981	1982	1983
M3 growth target range (%)	7–11	6–10	8–12	7–11
M3 growth outturn (%)	19.4	13	11.1	9.5
RPI inflation (%)	18	11.9	8.6	4.6
Unemployment (%)	6.9	9.8	10.9	11.6
Manufacturing output growth (%)	−8.7	−6.2	0	2.1

Table 6.1 Money supply targets vs. outturn and UK macroeconomic performance: 1980–83.

Note: Money supply targets and outturn run from February to April—e.g. 1980 refers to the period from February 1980 to April 1981.
Source: Goodhart (1986); UK Office for National Statistics.

Modelling monetary policy: active rule-based policy



Under inflation-targeting monetary policy framework, CB actively adjusts interest rates to keep inflation close to target

- E.g., this active central bank behaviour was witnessed between 1999–2022

Figure 6.1 Official central bank interest rates, monthly averages, %: 1999–2022.
Source: Bank of England; Board of Governors of the Federal Reserve System (US); Bank of Japan; European Central Bank (data accessed September 2022).

Modelling monetary policy: central bank preferences

Recall MR curve derivation in Ch. 3:

- CB minimizes its loss function $L = (y_t - y_e)^2 + \beta(\pi_t - \pi^T)^2$ subject to PC
 $\pi_1 = \pi_0 + \alpha(y_1 - y_e)$, which yields the $MR : (y_1 - y_e) = -\alpha\beta(\pi_1 - \pi^T)$
- Both β and α affect the chosen adjustment path (MR) after a shock
- (Fig 6.2 a, next slide) $\beta > 1 \rightarrow$ higher importance on π being away from π^T than y being away from $y_e \rightarrow$ larger output reduction when there is a π shock $\rightarrow$ flatter MR
- (Fig 6.2 b, next slide) $\alpha \uparrow$ (steeper PC) $\rightarrow \pi$ more responsive to $y \rightarrow$ able to cut π by more for a given cut in $y \rightarrow$ flatter MR

Modelling monetary policy: central bank preferences

β and/ or α higher $\rightarrow$ MR is flatter

a. Inflation averse policy maker

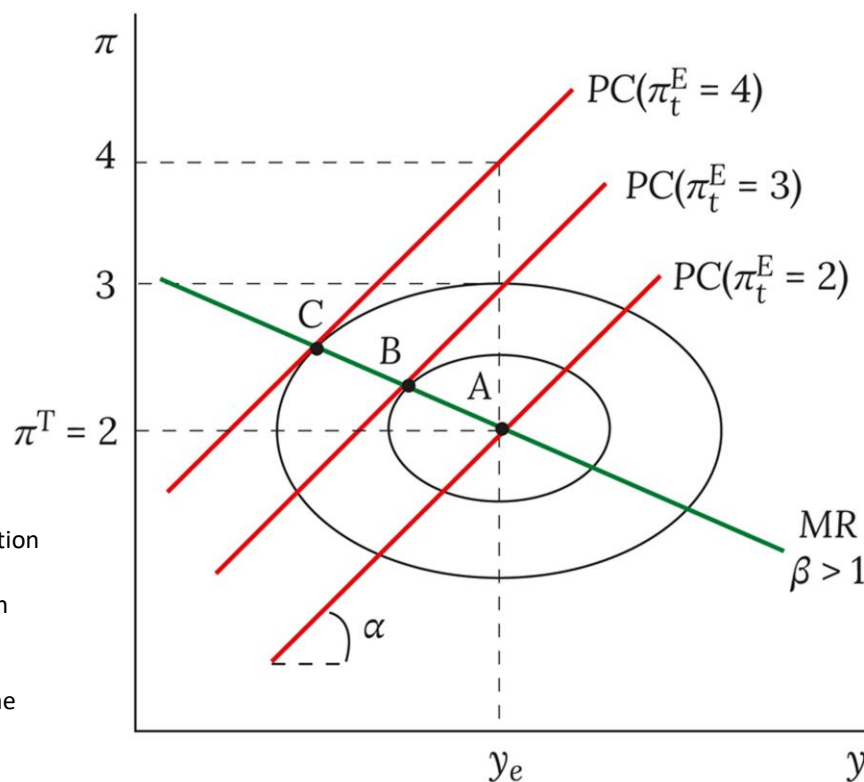
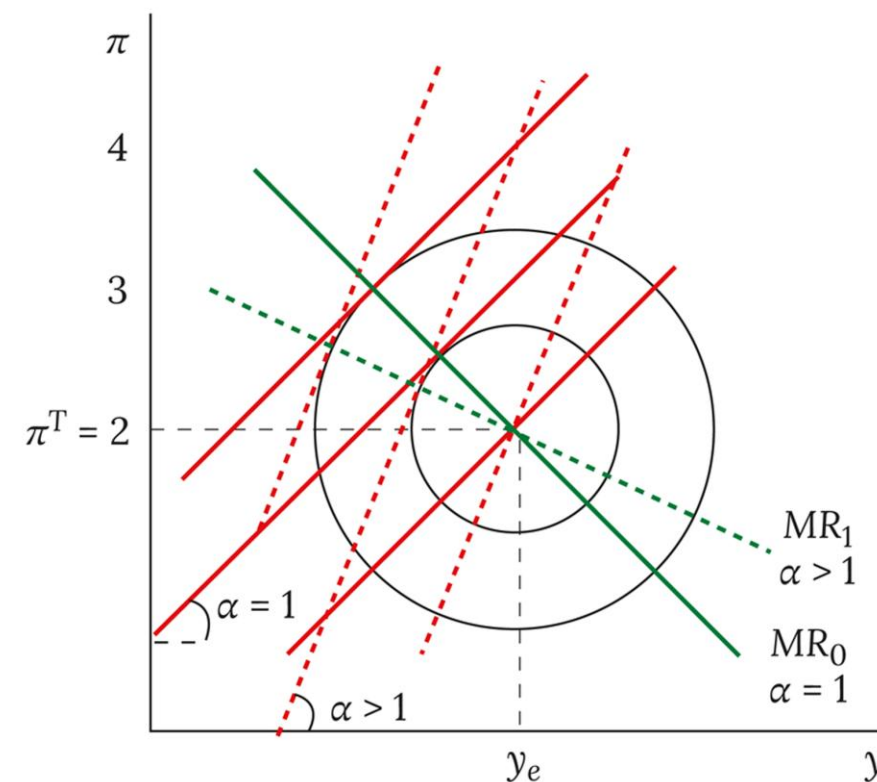


Figure 6.2 How varying the parameters in the 3-equation model affects the MR curve: a. Inflation-averse policy maker ('flattened' ellipses): flat MR line ($\beta > 1$) b. High responsiveness of inflation to output (steep Phillips curves): flat MR line ($\alpha > 1$).

Note: The angle marked α in the diagrams is in fact the angle whose tangent is α . We adopt this convention throughout.

b. High responsiveness of inflation to the output gap



Modelling monetary policy: central bank preferences

After π shock, the economy is at B

- Fig 6.3 a: with steeper PC , smaller cut in AD is best response (point D on MR_1)
- Fig 6.3 b: larger cut in AD given greater π -aversion

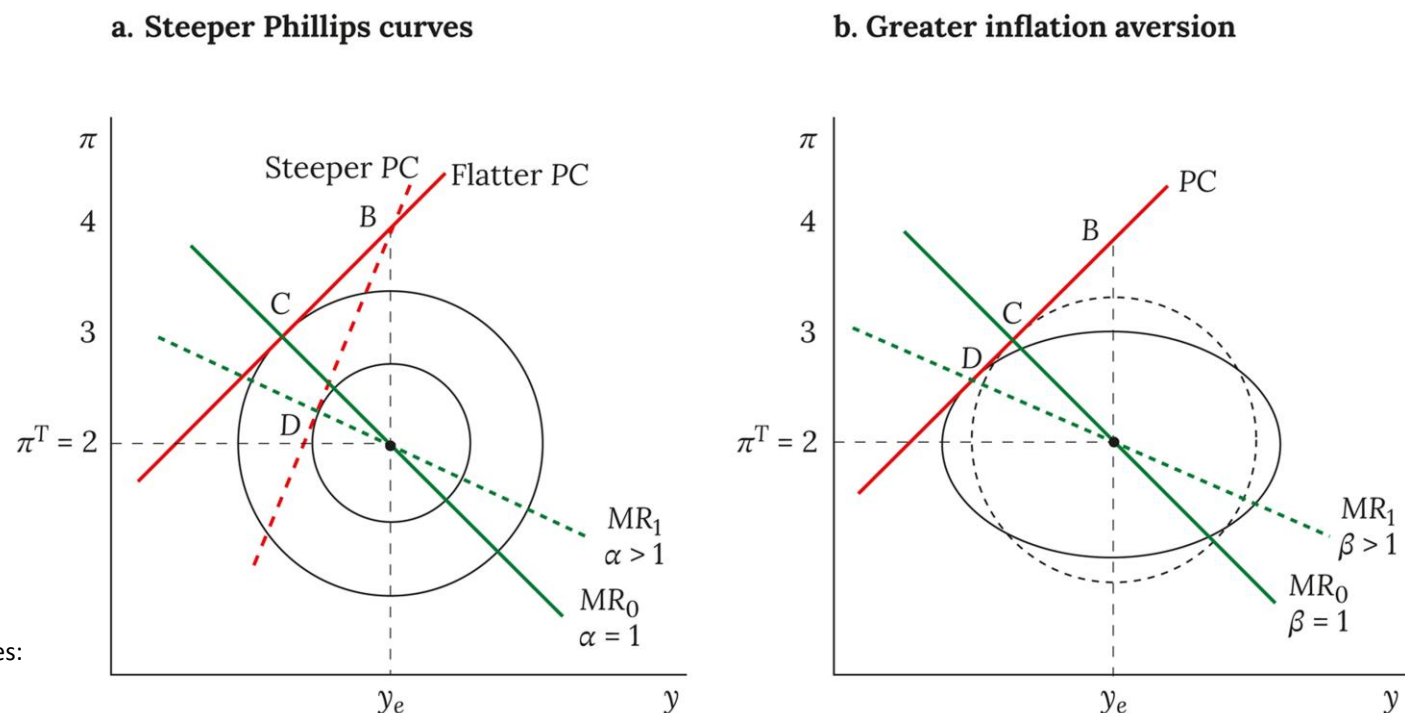
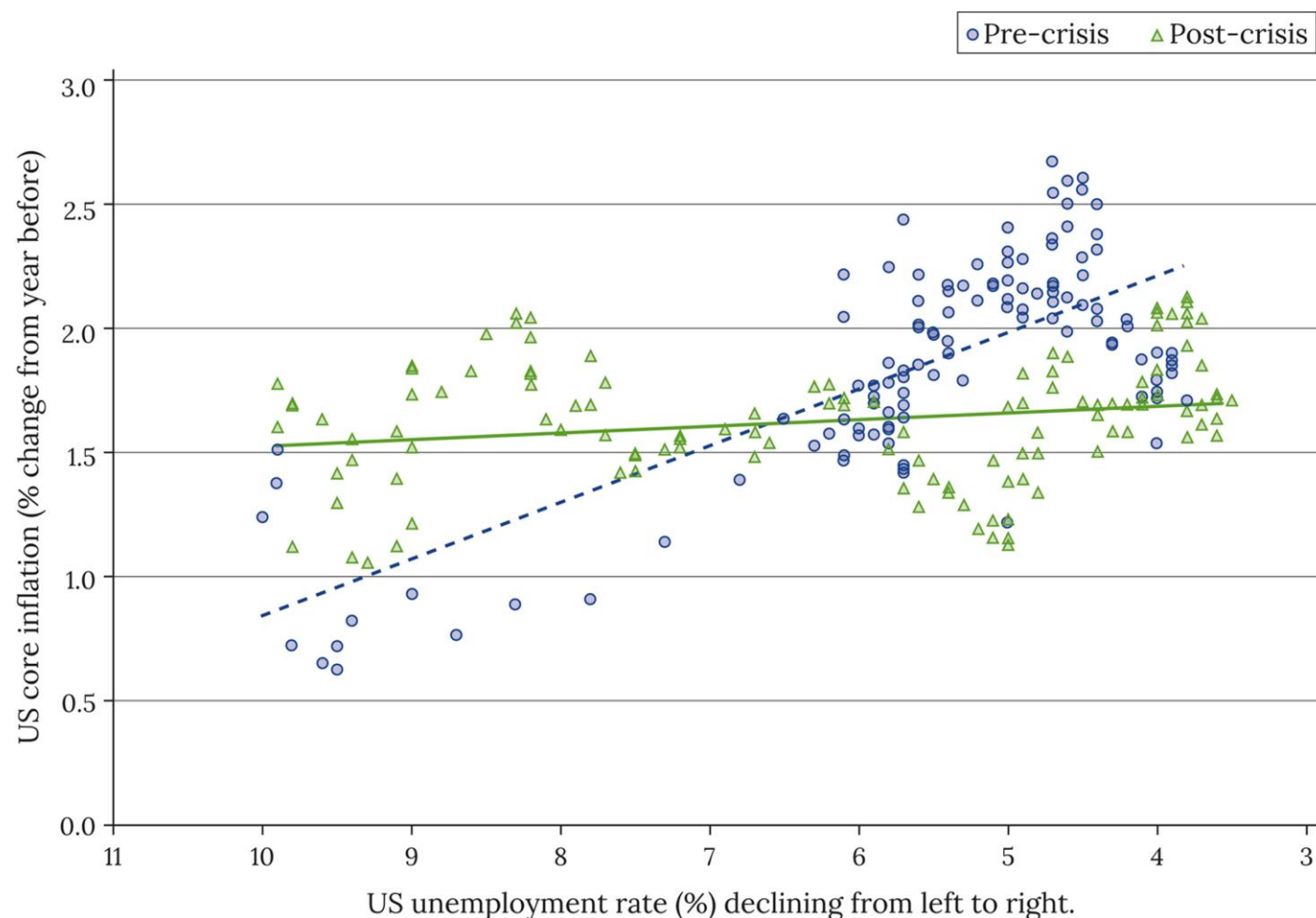


Figure 6.3 Comparing the response of the central bank in two cases: steeper Phillips curves and a more inflation-averse central bank.

Modelling monetary policy: flattening of PC



Inflation has become less sensitive to output over the years
→ PC flattens:

- Yet, when PC is plotted using wage growth, the flattening disappears
- Under successful inflation targeting, PC might not be observable in the out-turns for inflation and unemployment

Figure 6.4 Flattening of the Phillips curve in the United States from the second decade of inflation-targeting.

Notes: The chart plots core inflation (personal consumption expenditure excluding food and energy) against the unemployment rate, which declines from left to right. Pre-crisis data covers the period 2000–09, Post-crisis data covers the period 2010–19.

Source: FRED Economic Data (accessed November 2021).

Modelling monetary policy: sacrifice ratio

A way to capture the cost of disinflation is the sacrifice ratio: the **percentage point** rise in U for a 1% fall in π

Disinflation strategies:

- Cold turkey: more inflation aversion CB; achieve faster fall in π but with sharper rise in U
- Gradualist: U rises less, but disinflation process is longer

The shape of PC determines whether Cold turkey is more costly than Gradualist:

- If PC is linear: sacrifice ratios are the same under both strategies
- If PC is convex (flatter at higher unemployment): Cold turkey is more costly

Modelling monetary policy: Taylor rule

Taylor rule (1993) well described the Fed's historical interest rate behaviour:

- $r_0 - r_S = 0.5(\pi_0 - \pi^T) + 0.5(y_0 - y_e)$

Best response Taylor rule in our 3-equation model:

- Derivation:
 - Substitute $\pi_1 = \pi_0 + \alpha(y_1 - y_e)$ from PC into the MR : $(y_1 - y_e) = -\alpha\beta(\pi_1 - \pi^T)$, yielding $\pi_0 - \pi^T = -(\alpha + \frac{1}{\alpha\beta})(y_1 - y_e)$
 - Then substitute for $(y_1 - y_e) = -a(r_0 - r_S)$ from IS
 - We get the interest-rate rule (Taylor rule): $r_0 - r_S = \frac{1}{a(\alpha + \frac{1}{\alpha\beta})} (\pi_0 - \pi^T)$

Modelling monetary policy: Taylor rule

If $a = \alpha = \beta = 1$: $r_0 - r_s = 0.5 (\pi_0 - \pi^T)$

- If inflation is 1% above target, the CB needs to raise real interest rate by 0.5%
- By the Fisher equation, nominal interest rate(i) needs to raise by $1 + 0.5\%$ so that $r_0 = i_0 - \pi_1^E$ can be raised by 0.5%
 - Taylor principle: The need to raise i sufficiently to push up r , so that the CB's interest rate response is actually stabilizing

Optimal CB response is given by $r_0 - r_s = \frac{1}{a\left(\alpha + \frac{1}{\alpha\beta}\right)} (\pi_0 - \pi^T)$, thus:

- $\uparrow \beta \rightarrow$ High π aversion $\rightarrow$ respond to shock with larger rise in r
- $\uparrow \alpha \rightarrow PC$ steeper and MR flatter $\rightarrow$ smaller r response to shock
- $\uparrow a \rightarrow IS$ flatter (high r -sensitivity of AD) $\rightarrow$ smaller r response needed

Monetary policy transmission: transmission channels

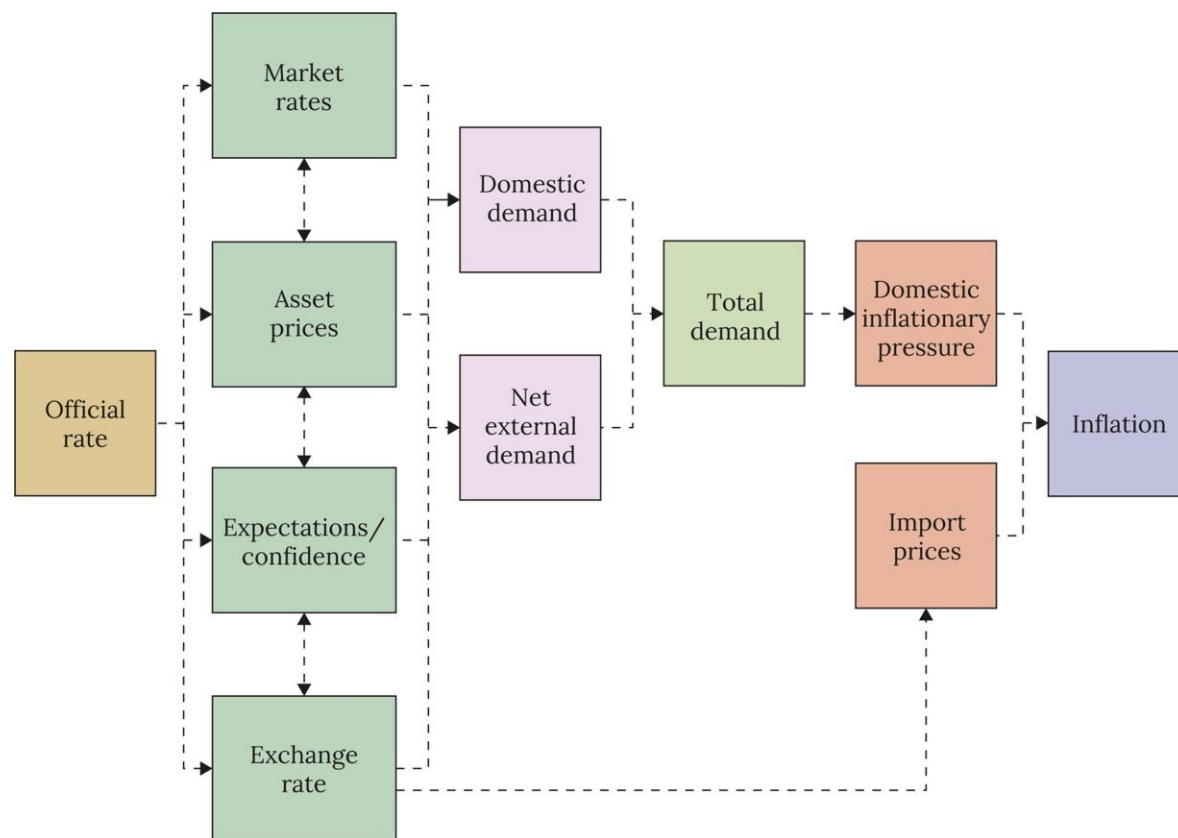


Figure 6.5 Monetary policy transmission mechanisms.
Source: Bank of England (accessed June 2012).

CB affects AD using i , working through 4 main channels:

- Market rates: $i \downarrow \rightarrow$ Borrowing desirable, saving less $\rightarrow AD \uparrow$
- Asset prices: $i \downarrow \rightarrow$ Asset price $\uparrow \rightarrow$ household wealth $\uparrow \rightarrow C \uparrow$ (via PIH)
- Expectations/confidence: $i \downarrow \rightarrow$ might show CB commitment to accommodative policy stance $\rightarrow AD \uparrow$
- Exchange rate channel: $i \rightarrow e \uparrow \rightarrow (X - M) \uparrow \rightarrow AD \uparrow$

Monetary policy transmission: Taylor rule in practice

Inflation-targeting CB around the world share similar objectives of price stability, full employment and economic growth, yet they have different mandates:

- Dual mandates on π and y : E.g., Federal Reserve
- Price stability as a priority and employment as secondary: E.g., Bank of England

Different CB mandates translated into different interest rate behaviour:

- A more general Taylor rule is: $i_t = \bar{i} + \gamma_1(\pi_t - \pi^T) + \gamma_2(y_t - y_e)$
 - $\bar{i}$ represents nominal interest rate
 - γ_1 and γ_2 depend on CB's mandates
 - By Taylor principle, γ_1 has to exceed one to increase real interest rate when there's an increase in inflation BoE and ECB react more strongly to inflation as opposed to the US Fed (see Castro, 2011).

Monetary policy transmission: Taylor rule in practice

All three CBs react to deviations in both inflation and output but to different degrees:

- BoE and ECB react more strongly to inflation as opposed to the US Fed
- Fed has a more balanced response given dual mandate

Central bank (time period)	Inflation gap coefficient	Output gap coefficient
Bank of England (1992–2007)	1.87 (4.89)	0.91 (2.8)
ECB (1999–2007)	2.77 (2.85)	1.99 (5.84)
Federal Reserve (1982–2007)	1.53 (5.18)	1.40 (2.77)

Table 6.3 Estimated coefficients for linear Taylor rules prior to the global financial crisis.

Note: T-statistics using robust standard errors are shown in brackets, all coefficients are significant at the 1% level.

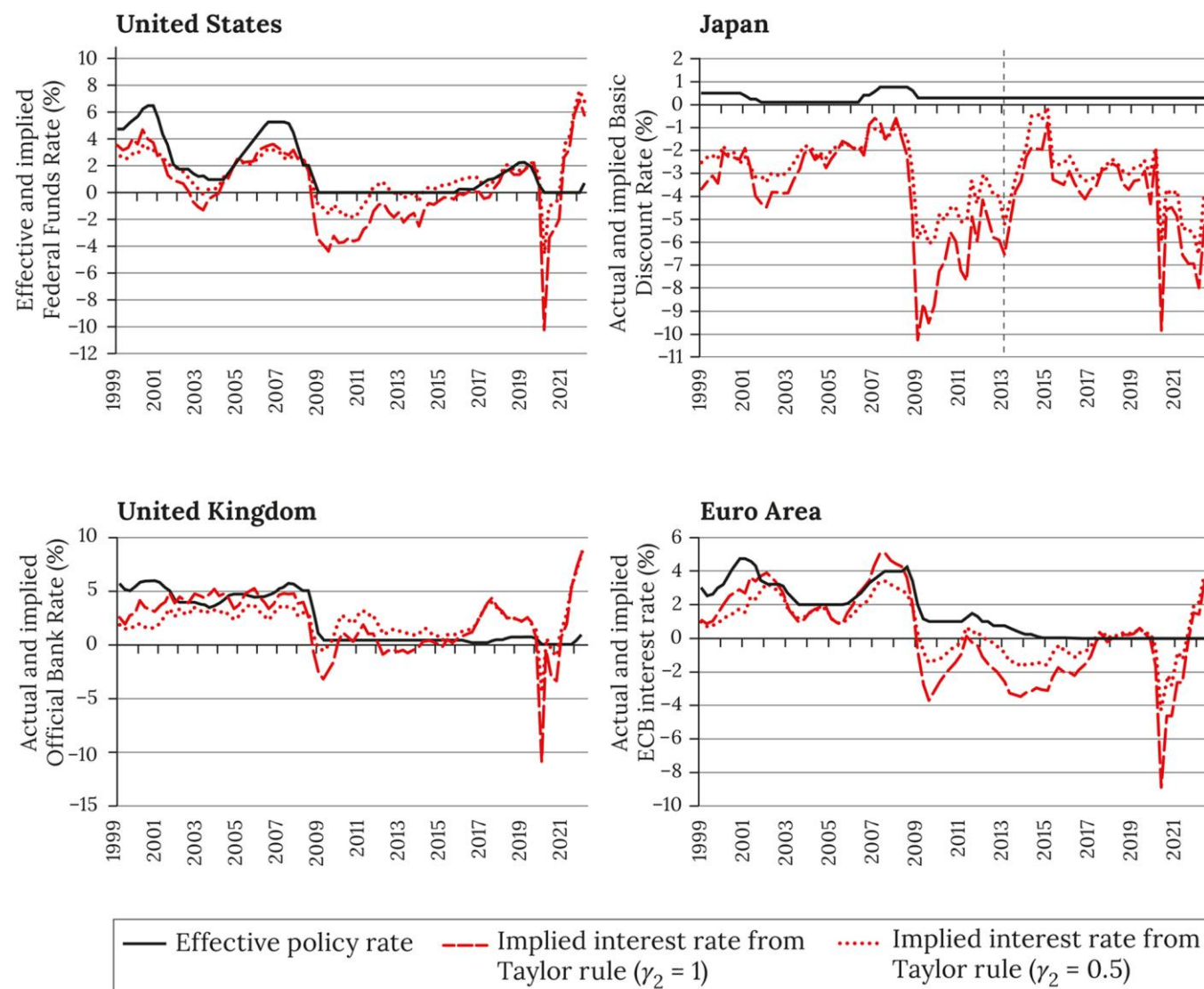
Source: Castro (2011).

Monetary policy transmission: asset price bubble

An asset price bubble occurs when financial market valuations become unrealistic

- For Dot-com boom in late 1990s and Mortgage boom in mid-2000s: CB chose not to prick the bubbles, but mop up after they burst
- Asset bubbles are undesirable but CB intervention to burst them depends on CB:
 - being able to identify bubbles before a financial crash occurs
 - refrain from identifying bubbles that do not exist
 - burst bubbles without causing excess damage to the wider economy
- Greenspan doctrine: CB is not able to fulfil these criteria
 - Even if CB can correctly identify bubbles, its main tool – the interest rate – is inadequate

Monetary policy transmission: zero lower bound



When ZLB is hit, the *TR* calls for negative nominal interest rates

- conventional monetary policy becomes ineffective for stabilization

Figure 6.6 Nominal interest rates implied by Taylor rules vs. observed central bank policy rates: 1999 Q1–2022 Q2.

Source: National statistical offices, national central banks, OECD, Eurostat (data accessed October 2022); Oxford Economics (October 2022). Note that the vertical line for 2013 Q1 for Japan indicates a break in the series.

Monetary policy transmission: QE transmission mechanism

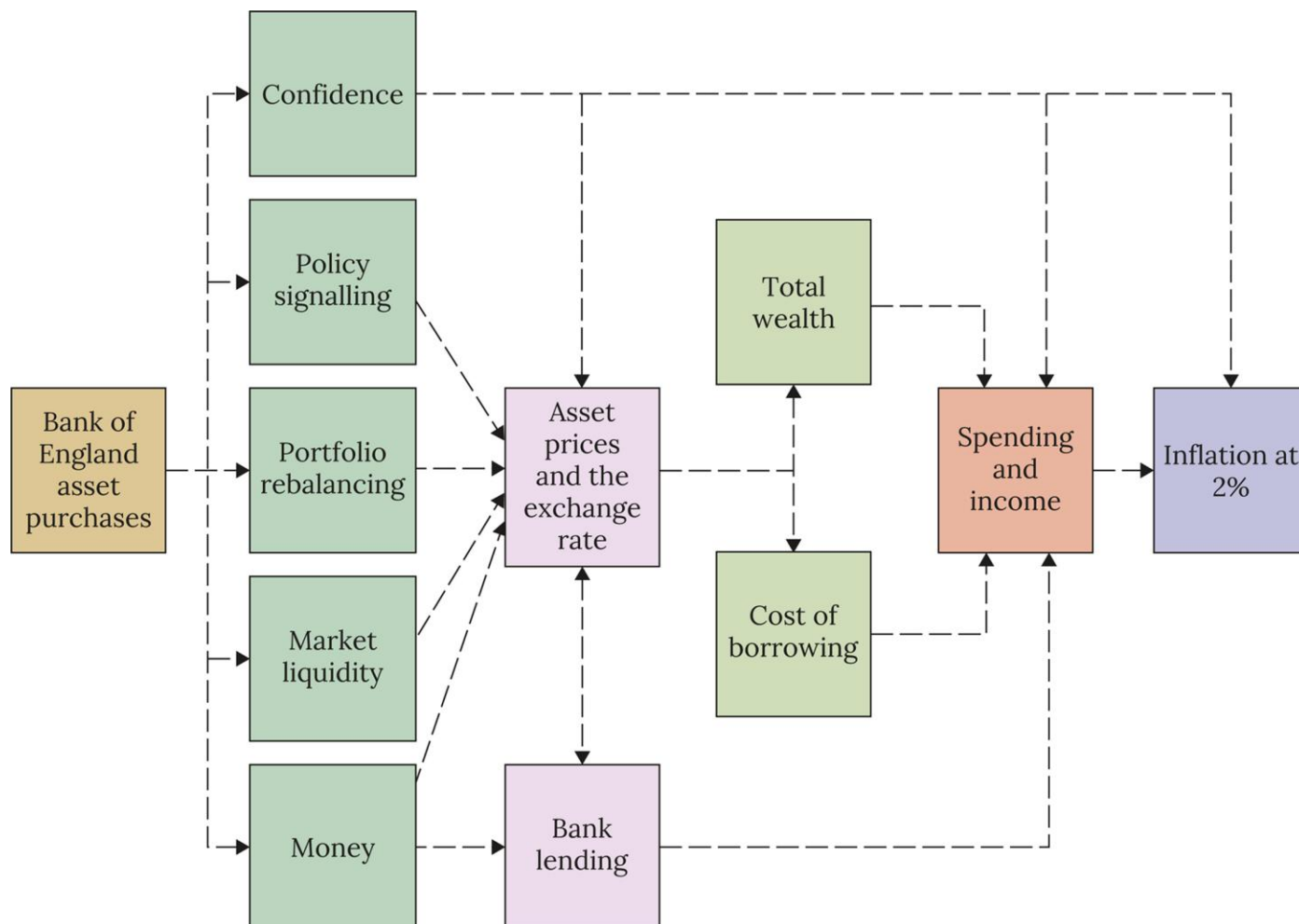


Figure 6.7 Quantitative easing transmission channels.

Source: Joyce et al. (2011).

Unconventional monetary policy: Quantitative Easing (QE)

- Targets long-term interest rates
- 5 Transmission channels of QE, as taken from the BoE Quarterly Bulletin (2011 Q3):
 - Confidence; policy signaling, portfolio rebalancing; market liquidity; money

Monetary policy transmission: QE transmission mechanism

Five transmission channels of QE from Fig 6.7:

- Confidence: directly affect asset prices
- Policy signaling: QE shows commitment to π^T , helping to anchor π^E and help prevent a deflation trap
- Portfolio rebalancing: Money from bonds sold to the CB can be used to buy other financial assets, which lowers yield and boosts demand
- Market liquidity: encourages trading and boosts liquidity during distress
- Money: more deposits into financial institutions may induce more lending

Measuring the impact of QE: difficult due to that absence of counterfactual; studies showed fall in government bond yields; QE was equivalent to a 1.5-3% point cut in the CB policy rate (Joyce et. al., 2011)

Monetary policy transmission: concerns in applying QE

Possible dangers of undertaking QE:

- CB independence and credibility ↓: future inflation-targeting becomes harder
- Inflation expectations ↑: QE might lead to higher future inflation, but higher π^E was necessary to prevent a deflation trap
- Financial stability: promotes search for yield, which can inflate asset bubbles
- Excess reserves: banks can make more loans → inflationary risk
- Pressure on commercial banks' profitability: QE flattens the yield curve → returns on long-term loans ↓ → commercial banks' profit ↓
- Exit from QE: the timing of quantitative tightening is crucial

Monetary policy transmission: forward guidance

CB transparency about the likely path of policy rates may help agents to form more accurate expectations of π and i :

- Forward guidance: communication by a central bank aimed at signaling the likely future path of policy rates
 - In an environment of near zero interest rates and a weak recovery, forward guidance is particularly useful as it signals CB's commitment to accommodative policy
 - However, given uncertainty the policymakers face, the feasibility of forward guidance is limited as the interest rate path is hard to judge

Monetary policy transmission: modifications after the pandemic

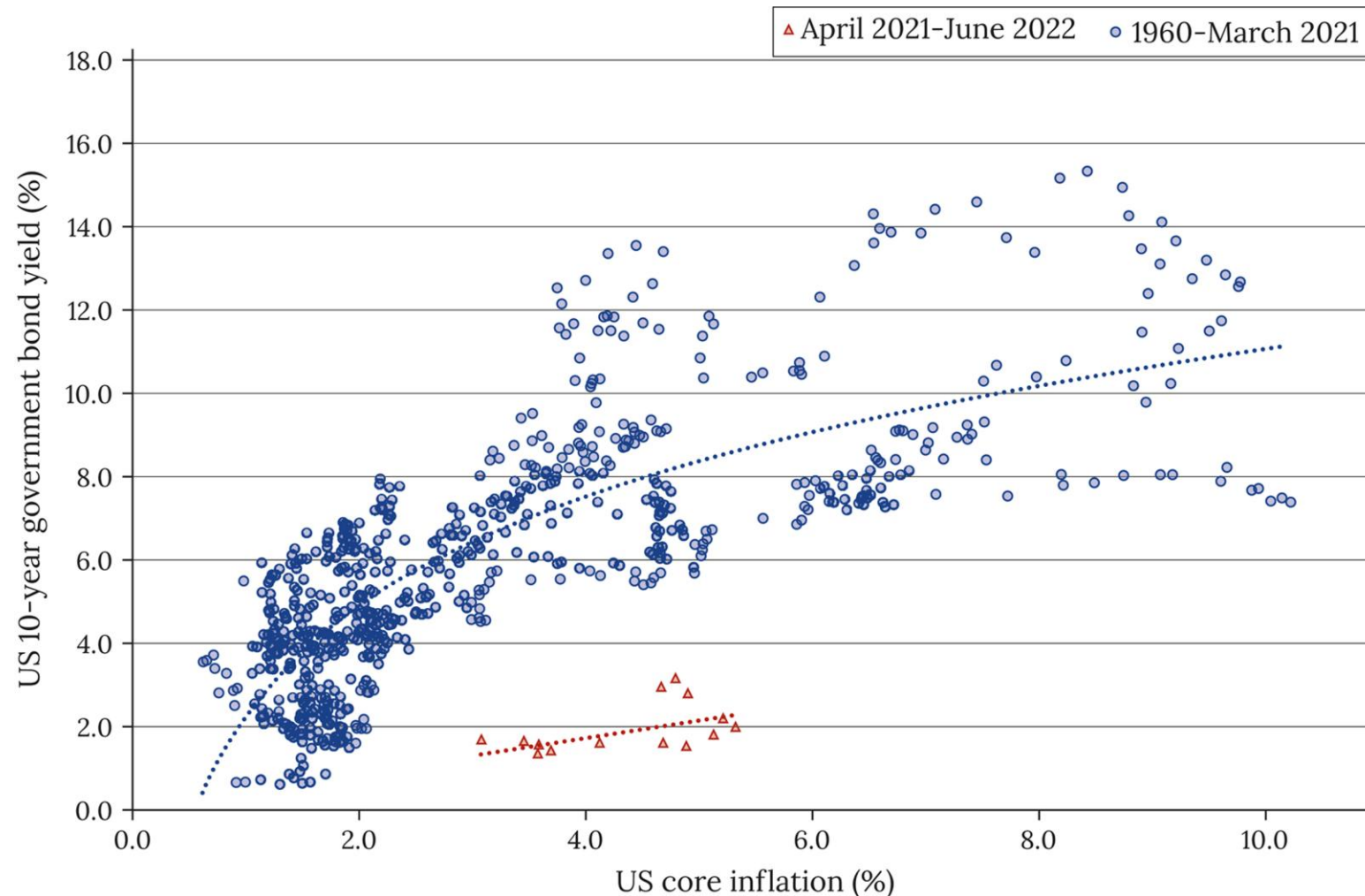


Figure 6.8 US inflation ('core' inflation) and 10-year US bond yields at monthly frequency (January 1960–June 2022).

Source: Extended from TS Lombard using data from FRED, accessed June 2022.

Prompted by the Covid-19, monetary policymakers modified their policy rules:

- Fed adopted average inflation targeting and this changed the market's interpretation of Fed policy

Monetary policy transmission: balancing CB objectives

With multiple responsibilities, CB sometimes face conflicting objectives:

- 2022 UK gilt crisis: CB voted for QT to tackle high inflation → then the UK government announced expansionary fiscal policy → large sell-off of UK government bonds (gilts) → CB aborted QT and announced QE for financial stability

Under excess reserves regime, CB can ensure financial stability without having to backtrack on inflation targets:

- 2022 UK: BoE's commitment to keep raising i^P to tackle inflation remained credible while it expanded QE. The size of QE can be varied without affecting the policy interest rate

Monetary policy transmission: inequality

QE impacts inequality through two channels: 'earnings' and 'asset price' channels

- Case: CB engages in QE under recession at ZLB
 - 'earnings' channel: inequality falls as lower income households benefit from increase in employment
 - 'asset price' channel: inequality increases as the lower interest rate raises the price of the assets held by the rich
- Evidence (Bartscher, et al. 2021):
 - Although income inequality is reduced, wealth inequality increases as White households benefit more than Black households
 - With higher wealth, the richer households increase consumption and have more access to finance
- Hence, it is important to take these channels into consideration when implementing monetary policy

Monetary policy transmission: inflation risk under QE

CB balance sheet expands with QE. The rising money supply may lead to rising inflation and even hyperinflation:

- Case: a spending boom with the models of Ch 1-5
 - Conditions for inflation: CB prevented from acting against rising inflation (fiscal dominance); commercial banks view the conditions for lending are met
 - Conditions for hyperinflation:
 - Empirical evidence: annual inflation rate rises to well over 100%.
 - Mechanism: once inflation rises to a high level for some time, the demand for money suddenly collapses and hyperinflation ensues
- Causal chain: households and firm spend such that $y > y_e \rightarrow$ commercial banks view loan requests as creditworthy $\rightarrow$ banks make more loans $\rightarrow$ CB doesn't raise interest rate to fight inflation due to political pressure $\rightarrow$ inflation rises and continue to rise
 - Unlikely to hold

Monetary policy transmission: quantity theory of money

Quantity Theory of Money: a causal model from the money supply to the price level and inflation

$$Py = MV \ (\blacktriangle)$$

- M : money supply, V : velocity of circulation of money, P : price level, y : output
- Convert $\blacktriangle$ to growth rates: $\frac{\Delta P}{P} + \frac{\Delta y}{y} \approx \frac{\Delta M}{M} + \frac{\Delta V}{V}$
 - Assuming $\Delta y = 0$ and a constant V , $\frac{\Delta P}{P} \approx \frac{\Delta M}{M} \rightarrow \pi \approx \frac{\Delta M}{M}$
- Wages and prices assumed to be perfectly flexible, so economy is always at MRE ($y = y_e$)
 - M pins down P ; growth of M pins down π

Monetary policy transmission: monetary-base multiplier

Monetary-base multiplier: a causal model from the monetary base (closely related to reserves) to the money supply

- Conditions for inflation increases as CB expands its balance sheet:
 - Households and firms hold a fixed ratio of banknotes to deposits (cd)
 - Commercial banks have a fixed reserve to deposit ratio
 - Velocity of circulation is fixed
 - full employment output level
- As a result, the increased supply of reserves leads to rising inflation

However, there is no fixed monetary-base multiplier in modern economy and banking system. Banks make lending decisions based on their business model

Summary

In this chapter, we have analyzed monetary policy further:

- MP evolved from M -targeting to π -targeting, where the CB optimizes and sets MP based on its preferences over y and π
 - CB displays Taylor Rule-behavior to set interest rates
- The best response MP (ie. MR) is affected by π -aversion(β) and the $PC(\alpha)$
 - Higher π -aversion $\rightarrow$ Flatter $MR \rightarrow$ More aggressive in raising r to dampen π , even if $U \uparrow$ a lot $\rightarrow$ ‘Cold turkey’, as opposed to ‘gradualist’
- Optimal Taylor Rule: best response r depends on the π -aversion(β), PC slope(α) and interest-sensitivity of AD (a)
 - CB around the world have different mandates and preferences

Summary

- Difficulty in identifying asset bubbles and using *MP* to burst them → ‘Mop-up’ approach before; but post-crisis, CBs now use macro-prudential policy
- Quantitative easing helped reduce long run i but may have its dangers
 - E.g., CB independence and credibility, higher inflation expectations and inequality issues
- Expansion of CB balance sheet may lead to higher inflation:
 - Models of Ch1-5: no direct causal connection between the growth of CB balance sheet (reserves) and inflation
 - Quantity Theory of Money and the monetary base multiplier: growth of reserves leads to inflation